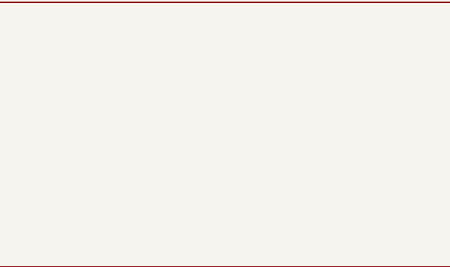


12 V / 100 A THREE-PHASE ALL-NMOS MOTOR DRIVER

DShot150/300/600/1200 via AM32 | STM32G071CBT6 + DRV8323R | 12x N-channel MOSFET, 2 per switch position

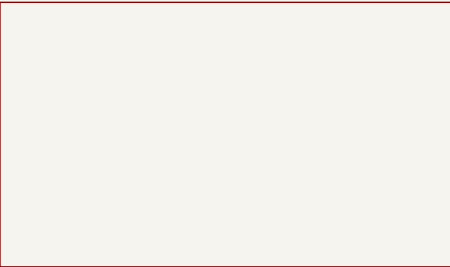
Rev A.1

Power Input



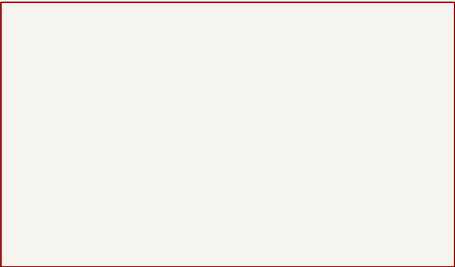
File: 01_power_in.kicad_sch

Gate Driver



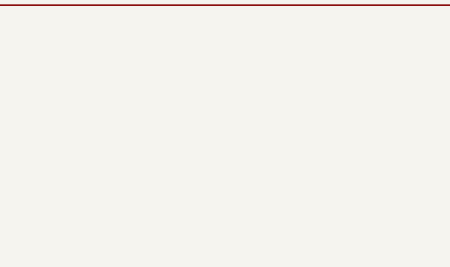
File: 02_gate_driver.kicad_sch

Power Stage



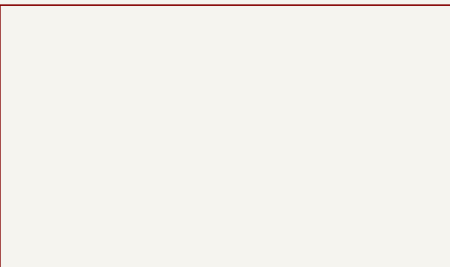
File: 03_power_stage.kicad_sch

MCU



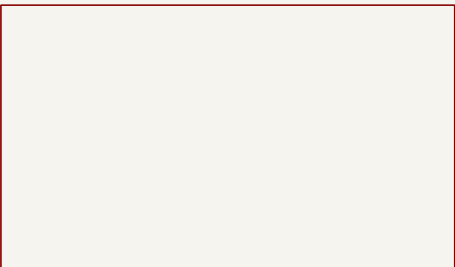
File: 04_mcu.kicad_sch

Sense



File: 05_sense.kicad_sch

Protection



File: 06_protection.kicad_sch

SHOOT-THROUGH is prevented in four independent layers:

- 1 TIM1 hardware dead-time generator, DEAD_TIME 40 = 625 ns
- 2 DRV8323R internal cross-conduction lockout + 100 ns
- 3 6x SN74LVC1G97 combinational interlock (sheet 06)
- 4 gate-source pulldowns + per-FET series gate resistors
- 0 nFAULT -> PB12/TIM1_BKIN kills all 6 outputs in hardware

Full budget and derivation: docs/shoot-through.md

100 A continuous / 150 A burst at 12.6 V = ~19 W / ~40 W

dissipated. Needs forced air -- see docs/power-budget.md.

Firmware: AM32, HARDWARE_GROUP_G0_A + NO_PA11_PA12_REMAP.

See firmware/README.md for the target block and the

DRV8323RS SPI init that AM32 does not ship.

STM32G071CBT6 + DRV8323R | 12x TPHP8504PL | DShot via AM32

Top sheet - hierarchy overview

Lucas Poulos

Sheet: /

File: dshot-esc-100a.kicad_sch

Title: 12V 100A Three-Phase All-NMOS Motor Driver

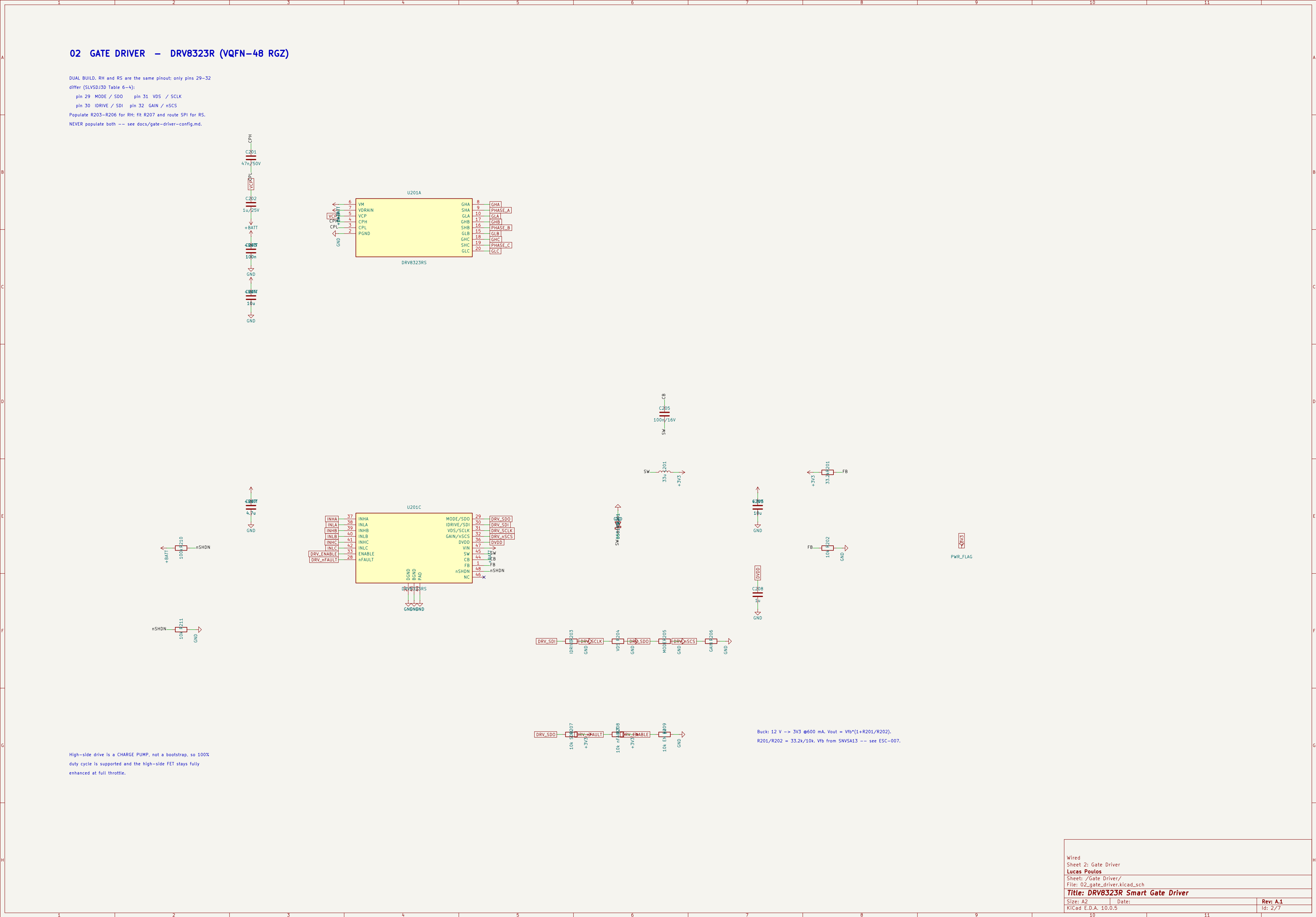
Size: A2

Date:

Rev: A.1

KiCad E.D.A. 10.0.5

Id: 1/7

[illegible][illegible]

02 GATE DRIVER — DRV8323R (VQFN-48 RGZ)

DUAL BUILD. RH and RS are the same pinout; only pins 29-32 differ (SLVSD/3D Table 6-4):

pin 29 MODE / SDO pin 31 VDS / SCLK
pin 30 IDRIVE / SDI pin 32 GAIN / nSCS

Populate R203-R206 for RH; fit R207 and route SPI for RS.
NEVER populate both -- see docs/gate-driver-config.md.

High-side drive is a CHARGE PUMP, not a bootstrap, so 100% duty cycle is supported and the high-side FET stays fully enhanced at full throttle.

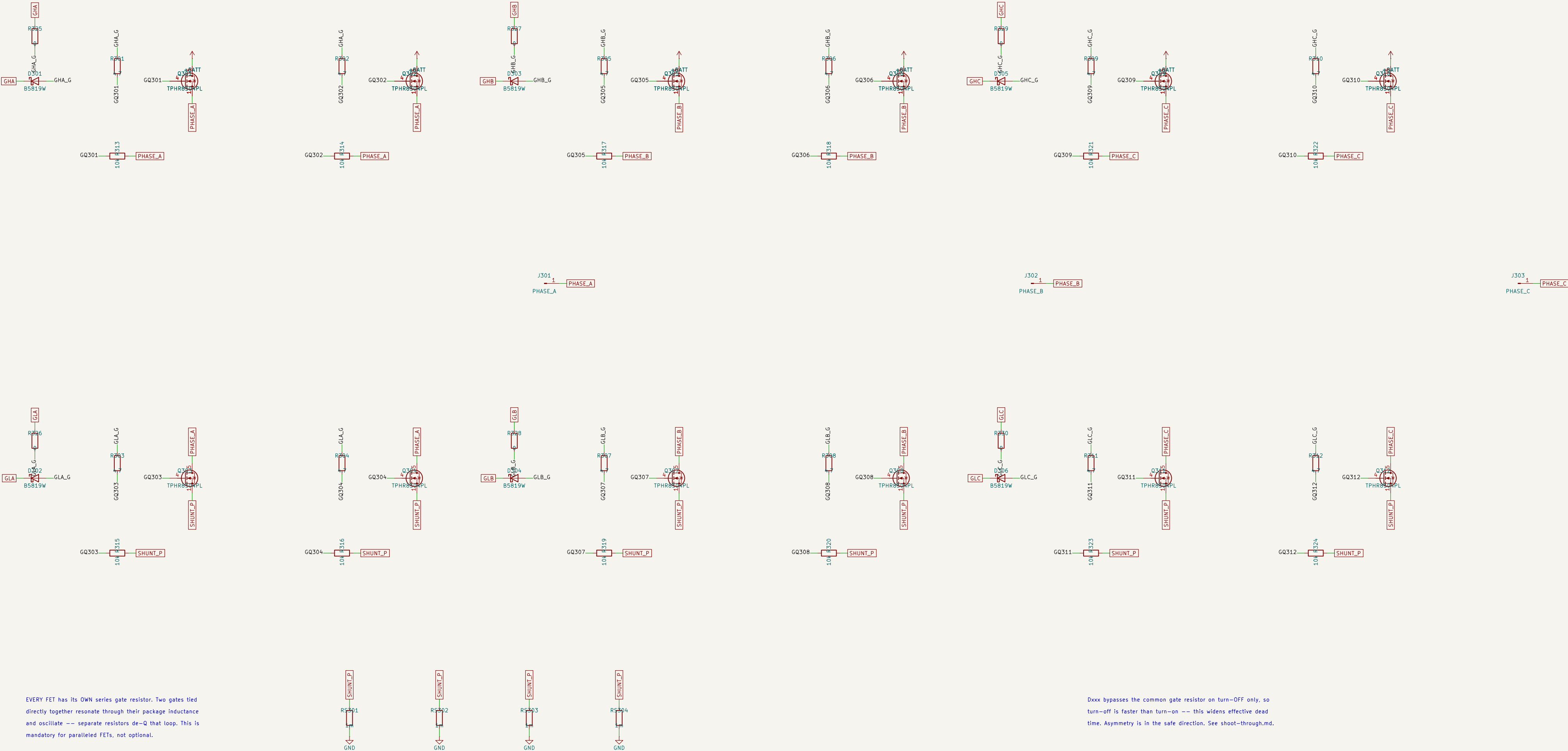
Buck: 12 V -> 3V3 @600 mA. $V_{out} = V_{fb} \cdot (1 + R201/R202)$.
 $R201/R202 = 33.2k/10k$, V_{fb} from SNVSA13 -- see ESC-007.

Wired
Sheet 2: Gate Driver
Lucas Poulos
Sheet: /Gate Driver/
File: 02_gate_driver.kicad_sch
Title: DRV8323R Smart Gate Driver
Size: A2 Date: Rev: A.1
KiCad E.D.A. 10.0.5 Id: 2/7

[illegible]

03 POWER STAGE – 12x N–channel, 2 per switch position

TPHR8504PL 40 V / 0.85 mOhm max, 40 V on a 12.6 V bus is deliberate: a 150 A edge into ~5 nH of loop inductance adds 7.5 V of overshoot and real layouts do worse.

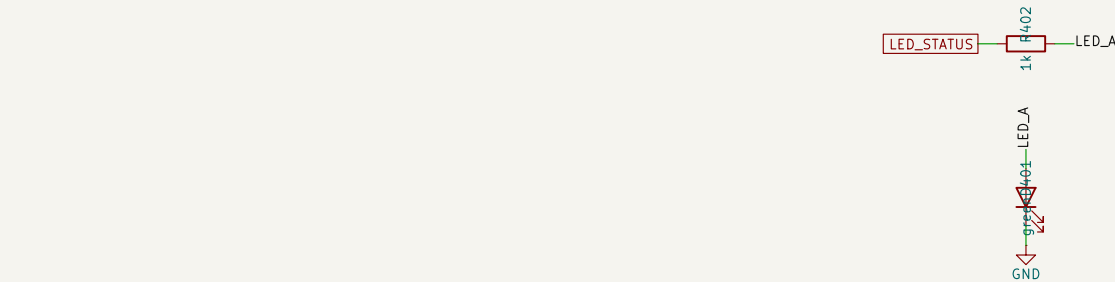
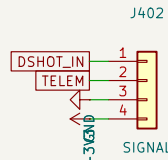
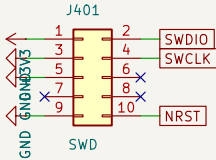
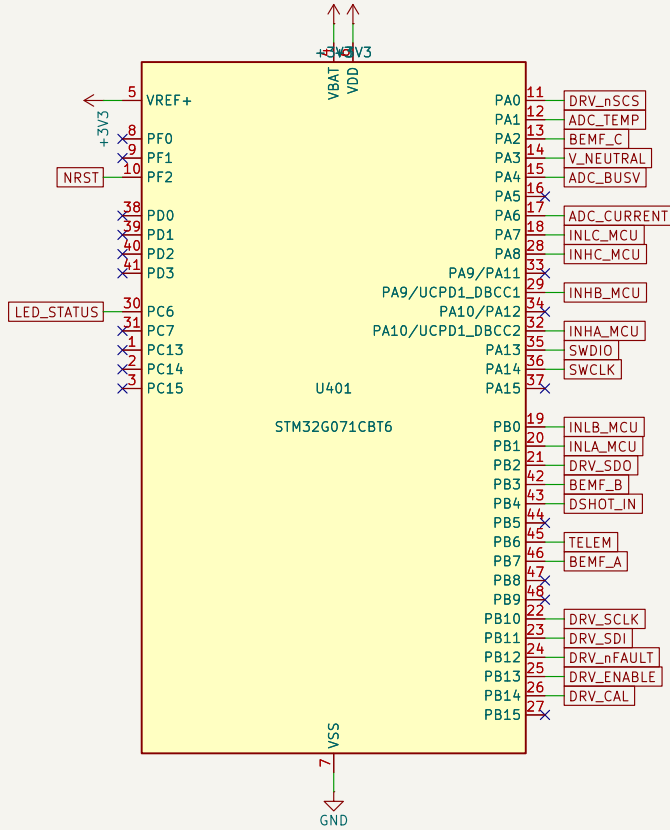
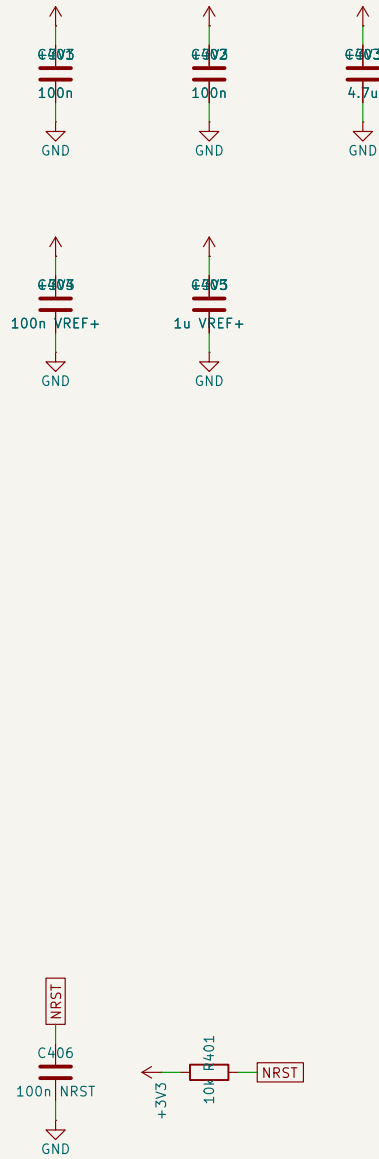


Dxxx bypasses the common gate resistor on turn-OFF only, so turn-off is faster than turn-on -- this widens effective dead time. Asymmetry is in the safe direction. See shoot-through.md.

RS301-304: 4x 1 mOhm 2512 = 0.25 mOhm, 12 W rating, 2.5 W dissipated at 100 A. Hottest discrete on the board.

04 MCU – STM32G071CBT6 (LQFP–48)

AM32 target: `HARDWARE_GROUP_G0_A + NO_PA11_PA12_REMAP`.
On the 48–pin package PA9/PA10 are DEDICATED pins 29/32;
pins 33/34 are PA11/PA12 and are left unconnected. AM32
remaps by default, which is correct for 28/32–pin parts and
WRONG here –– hence the define. See docs/pinout.md.



nFAULT lands on PB12 = TIM1_BKIN so a driver fault kills all
six PWM outputs in `HARDWARE`. See `ESC–003`.

Symbol is KiCad's `STM32G081CBTx` copied and renamed: the

G081CB is the same die and bonding as the G071CB (DS12232
covers both) and differs only by the AES block, which has
no pins. Avoids hand–transcribing 48 pins.

05 SENSE — back-EMF, bus voltage, current, temperature

AM32 multiplexes ONE comparator (COMP2) across the three phases. Phase dividers drive COMP2_INM; the virtual neutral drives COMP2_INP on PA3. R507–509 + R510 synthesise the star point, which an unterminated wye or delta motor never brings out. Divider ratio 11:1 matches the phase dividers exactly -- if they differ, zero-crossing detection is biased and timing drifts with throttle.

The schematic illustrates the sense circuitry for a motor control system. It includes three phase dividers (R501-R503) connected to PHASE_A, PHASE_B, and PHASE_C, which are then connected to BEMF_A, BEMF_B, and BEMF_C respectively. These signals are further processed by comparators (C504-C506) and a multiplexer (U201B). The U201B component has pins labeled SHUNT_P, SPA, SNA, SPB, SNB, SPC, SNC, SQA, SOA, SOB, SOC, VREF, CAL, AGND, and ADC_CURRENT. The output of the multiplexer is connected to the ADC_CURRENT pin. Additionally, there are three phase dividers (R507-R509) connected to PHASE_A, PHASE_B, and PHASE_C, which are then connected to V_NEUTRAL. A fourth divider (R510) connects V_NEUTRAL to GND. The bus voltage is measured via a shunt resistor (R511) connected to +BATT and GND, with the signal going to ADC_BUSV. The bus voltage is also connected to a capacitor (C501) and GND. The temperature sensor (T501) is connected to +3V3 and GND, with the signal going to ADC_TEMP. The temperature sensor is also connected to a capacitor (C502) and GND. The output of the temperature sensor is connected to the ADC_TEMP pin.

Only CSA A is used (AM32 reads total bus current). SPB/SNB and SPC/SNC are tied to PGND per SLVSDJ3D Table 6-4: an unused CSA input must be tied, not left floating. Three separate shunts would enable per-phase FDC later at 3x the shunt dissipation -- deferred, not designed out.

Wired
Sheet 5: Sense
Lucas Poulos
Sheet: /Sense/
File: 05_sense.kicad_sch
Title: Back-EMF, Current and Voltage Sensing
Size: A2 Date: Rev: A.1
KiCad E.D.A. 10.0.5 Id: 5/7

05 SENSE — back-EMF, bus voltage, current, temperature

AM32 multiplexes ONE comparator (COMP2) across the three phases. Phase dividers drive COMP2_INM; the virtual neutral drives COMP2_INP on PA3. R507-509 + R510 synthesise the star point, which an unterminated wye or delta motor never brings out. Divider ratio 11:1 matches the phase dividers exactly -- if they differ, zero-crossing detection is biased and timing drifts with throttle.



05 SENSE — back-EMF, bus voltage, current, temperature

AM32 multiplexes ONE comparator (COMP2) across the three phases. Phase dividers drive COMP2_INM; the virtual neutral drives COMP2_INP on PA3. R507–509 + R510 synthesise the star point, which an unterminated wye or delta motor never brings out. Divider ratio 11:1 matches the phase dividers exactly -- if they differ, zero-crossing detection is biased and timing drifts with throttle.

The schematic illustrates the sense circuitry for a motor controller. It includes three phase dividers (PHASE_A/B/C to BEMF_A/B/C) using resistors R501-R506. A virtual neutral divider (PHASE_A/B/C to V_NEUTRAL) uses resistors R507-R510. Temperature sensing is implemented with a network of resistors (R513-R514) and capacitors (C502) connected to the ADC_TEMP input. Bus voltage sensing is achieved through resistors (R511-R512) and capacitor (C501) connected to the ADC_BUSV input. The U201B DRV8323RS IC is shown with its pins connected to the sense signals and ground.

U201B
DRV8323RS

SHUNT_P 11 SPA 25 SDA 25
GND 12 SNA 26 SDB 26
GND 14 SPB 23 SOC 23
GND 13 SNB 26 VREF 26
GND 21 SPC 34 CAL 34
GND 22 SNC 35 AGND 35

ADC_CURRENT
DRV_CAL
GND

+3V3
+BATT
GND

ADC_TEMP
ADC_BUSV

C501
C502

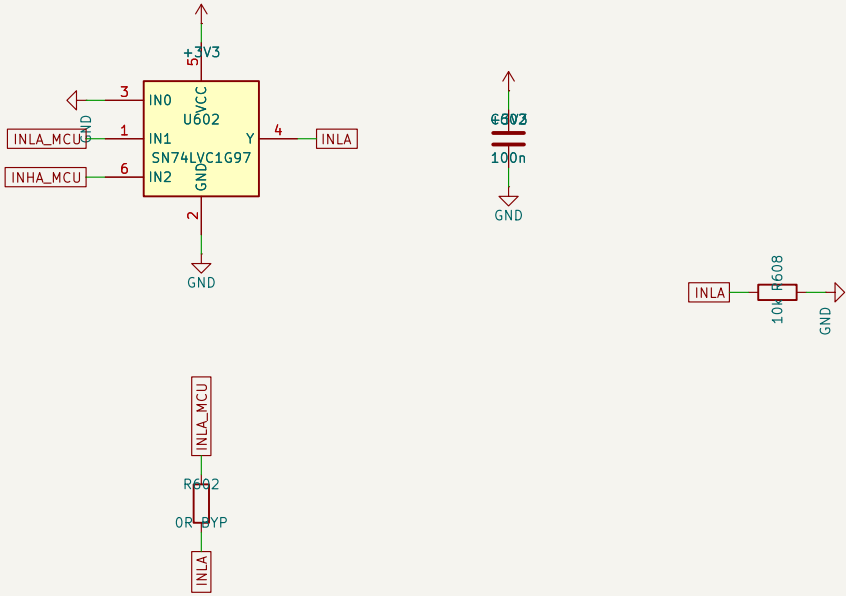
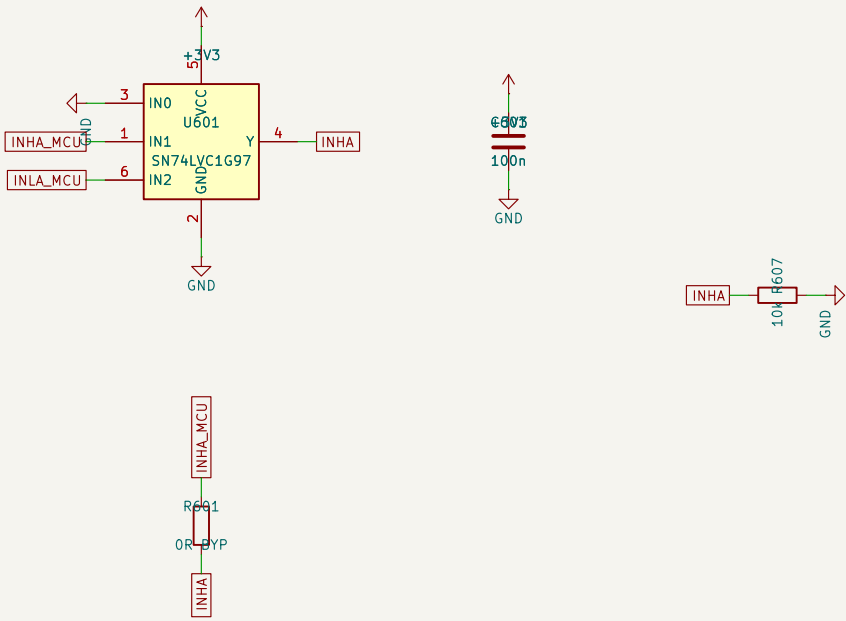
VREF

Only CSA A is used (AM32 reads total bus current). SPB/SNB and SPC/SNC are tied to PGND per SLVSDJ3D Table 6-4: an unused CSA input must be tied, not left floating. Three separate shunts would enable per-phase FOC later at 3x the shunt dissipation -- deferred, not designed out.

Wired
Sheet 5: Sense
Lucas Poulos
Sheet: /Sense/
File: 05_sense.kicad_sch
Title: Back-EMF, Current and Voltage Sensing
Size: A2 Date: Rev: A.1
KiCad E.D.A. 10.0.5 Id: 5/7

06 SHOOT-THROUGH INTERLOCK (layer 3 of 4)

SN74LVC1G97 wired per datasheet SCES416N Figure 6.
'2-Input AND Gate With One Inverted Input':
In0 (pin 3) = GND -> Y = In1 AND NOT In2
high side: In1=INHx_MCU, In2=INLx_MCU
low side: In1=INLx_MCU, In2=INHx_MCU
Both outputs cannot be high at once for ANY input pattern.
This is combinational -- no state to corrupt.



Largely redundant with the DRV8323R's own interlock, and

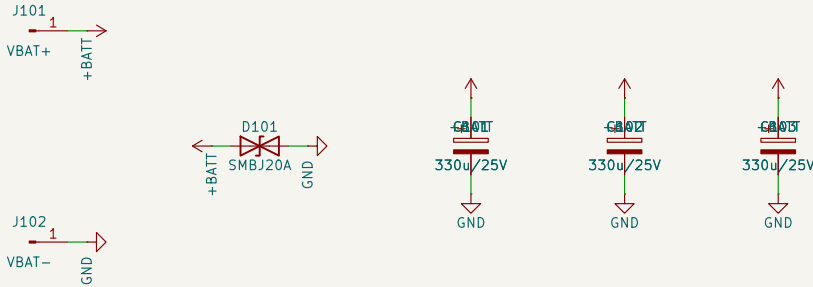
here anyway for the two cases it does not cover: the window before ENABLE+SPI config, and a firmware bug selecting 3x or 1x PWM mode (where INLx stops meaning 'low side on').

R601-606 are OR bypass links for characterisation. Fitting a bypass DEFEATS layer 3 -- do not ship a board with them in.

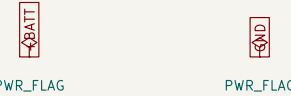
R607-612 pull the driver inputs down, so a tri-stated or unpowered MCU commands 'off' rather than 'undefined'.

01 POWER INPUT + BULK

3S LiPo, 9.0–12.6 V, 100 A continuous / 150 A burst.
D101 SMBJ20A: 20 V standoff clears 4S (16.8 V) and clamps
at 32.4 V, below the 40 V FET rating.



Reverse-polarity protection is deliberately ABSENT: an ideal-diode stage at 100 A costs ~0.5 mΩm and ~5 W plus its own gate drive. Mitigation is the polarised XT90. See ESC-006.



Ripple budget ~50 A RMS at 100 A: C101–103 carry ~8.4 A, the 24x 10µF MLCC array carries the rest. See ESC-001 -- a low-ESR cap at the battery connector on SHORT leads is REQUIRED: lead inductance dominates, not the on-board bank.

